Chapters

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Intro.

In this offshoot from the main document, I will go a bit deeper into chess.
(main document: https://github.com/MadHatter-Auditor/vortex-anapole-confinement/blob/main/Base-12%20Microcosm.pdf)
I will use the Greek Duodecimal System in this document. For more on that I have made the document GDS that further elaborates on it. https://github.com/MadHatter-Auditor/vortex-anapole-confinement/blob/main/GDS.pdf. The name **DSPH** stands for Duodecimal Symmetric Phase Domain.

Chess and the mystery of the freemasons.



The freemasons are shrouded in mystery and occultism. Like wizards and wise men whom hold the secrets of the universe. Actually, I wasn't even thinking that much about it until I saw the traditional tiles shaped like a chessboard. Some straight and some at an angle as shown in that exact picture to the right.

Knowing the club is heavily into geometry and esoterism it would seem likely they project deeper meanings in the obvious. Since it is common belief that freemasons love chess. It is a bit of a chicken or the egg story considering if that popular belief is formed from the tiles or that the tiles are inspired on the deeper meanings of chess.

Duodecimal Chess grid.

Now a chessboard has 8 by 8 tiles. So, let's try to divide them into a base.
9/8=1.125... okay so that gives me no clean number. Let's try Base-10...
10/8=1. 25.. okay a little bit better but what if we take a Base-12?
12/8=1. 5.. this works! Because 120/8=15. That is now the same as the rest. Because 8x15°=120°

To the right I have written down in both denary as in duodecimal. Here we can also clearly see how changing a base can change the patterns visible.

15	30	45	60	75	90	105	120
30	60	90	120	150	180	210	240
45	90	135	180	225	270	315	360
60	120	180	240	300	360	420	480
75	150	225	300	375	450	525	600
90	180	270	360	450	540	630	720
105	210	315	420	525	630	735	840
120	240	360	480	600	720	840	960

In the duodecimal grid we can see a clear symmetry in the quarters and a domination of the twelvefold numbers in green. But rather than in the denary grid where whole and halve units were dominating the board. We see more depth with duodecimal. It should be noted that I went no further than 100 in duodecimal using module-144.

13	26	39	50	63	76	89	α0
26	50	76	α0	6	30	56	80
39	76	ω3	30	69	α6	23	60
50	α0	30	80	10	60	ω0	40
63	6	69	10	73	16	79	20
76	30	α6	60	16	90	46	0
89	56	23	ω0	79	46	13	α0
α0	80	60	40	20	0	α0	80

I could have done the same for the above one but that would have made no difference in that case since using module-100 would have resulted in the same patterns.

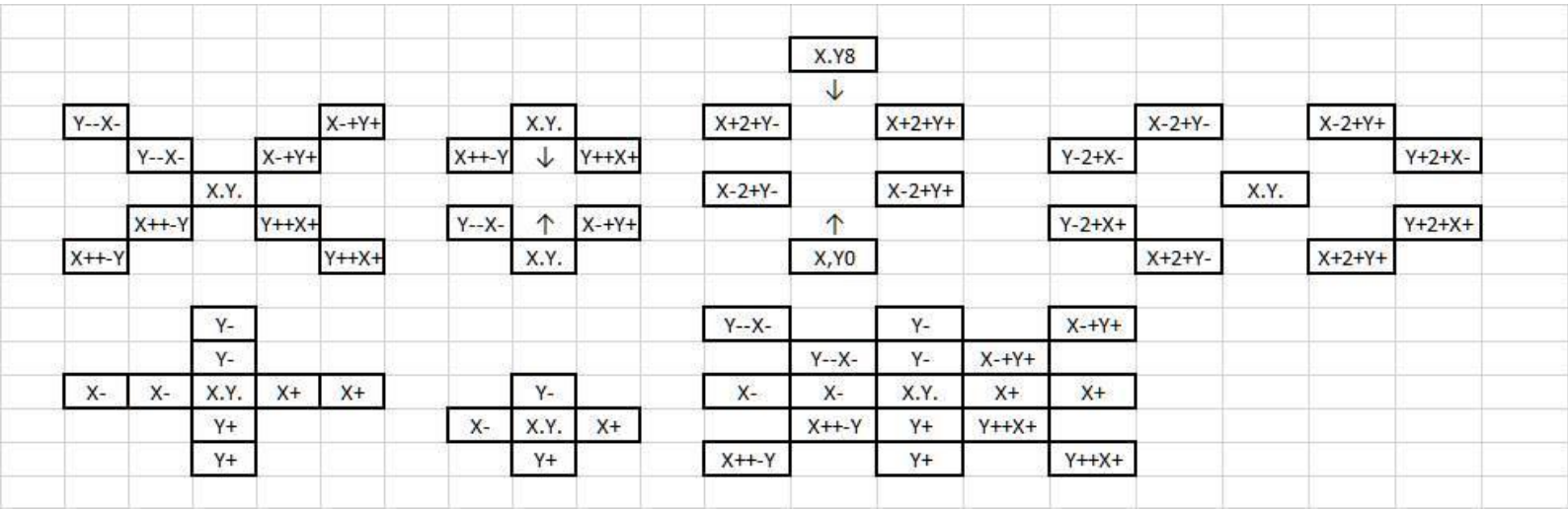
Duodecimal Chess in degrees.

		15°	30°	45°	60°	75°	90°	105°	120°
15°	Y8	13	26	39	50	63	76	89	α0
30°	Y7	26	50	76	α0	6	30	56	80
45°	Y6	39	76	ω3	30	69	α6	23	60
60°	Y5	50	α0	30	80	10	60	ω0	40
75°	Y4	63	6	69	10	73	16	79	20
90°	Y3	76	30	α6	60	16	90	46	0
105°	Y2	89	56	23	ω0	79	46	13	α0
120°	Y1	α0	80	60	40	20	0	α0	80
		X1	X2	X3	X4	X5	X6	X7	X8

Here I have also put the axis and degrees per row. One of the easier things to calculate now are positions. Because the grid is now clearly divided in 4 parts with identical patterns. Since the rows on both axis's determine how the steps are calculated it is important to name them. Normally chess is from A to H and 1 to 8, to map it accordingly I have changed it to coordinates. Interesting to note is that because the position defines also the steps it is also introducing a sense of local laws and definitions to the grid. It is not that every step is 15 although the whole grid is built on 15. The steps on row X1,Y1 are different than rows X5,Y5.

Now let's also have a look at the chess pieces.

Here I have mapped the steps possible and how the pieces behave. Interesting to note is that the position defines also the steps. This introduces a sense of local laws and definitions to the grid.



Feynman's model.

The **Feynman Checkerboard Model** is an elegant thought experiment from the 1940s. The legendary physicist Richard Feynman created it to demonstrate that the strange, complex laws of quantum mechanics can be replicated using a very simple game on a chessboard.
He used the board to calculate the movement of an **electron** (a fundamental building block of electricity).

How Feynman's Checkerboard Works
Imagine an infinite chessboard. Feynman established that the axes of the board represent two things:

- **The horizontal axis** represents **Space** (where the particle is located).
- **The vertical axis** represents **Time** (every row upward is one step forward in time).

Now, place a checker piece (the electron) on the bottom square. This electron must move upward (forward in time) and is only allowed to jump **diagonally left** or **diagonally right** with each time step. This means the particle is constantly moving at the speed of light.

The Quantum Rules of the Game
If this were a standard model, the particle would simply wander upward in a random path (a random walk). However, because this is quantum mechanics, Feynman added a mathematical "clock":

1. **Moving straight changes nothing:** As long as the particle keeps jumping in the same direction (for example, three times in a row diagonally right), its "phase" (energy state) remains unchanged.
2. **The 90-degree turn:** Every time the particle changes direction (from left to right, or right to left), it receives a "mathematical kick." Feynman dictated that the particle's internal quantum clock **rotates exactly 90 degrees clockwise** with each turn. Mathematically, you multiply its state by the complex number $-\mathbf{i}$.

Feynman's reasoning
In the quantum world, a particle can take *all possible routes simultaneously* (Feynman's famous *path integrals*).

If you want to know the probability of the electron landing on a specific square at the top of the board after 10 seconds, you must add up the outcomes of **all possible routes** to that square. Because one route might have 2 turns (90° + 90° = 180°, making it negative) and another might have 4 turns, the routes either constructively reinforce or completely cancel each other out (quantum interference).

When you play this out mathematically to infinity on the chessboard, it yields *exactly* the **Dirac Equation**. This is the official, Nobel Prize-winning formula that describes how electrons behave according to the theory of relativity.

This visual representation of the Dirac Equation displays a grid of dots where **the horizontal axis (x) represents space** and **the vertical axis (t) represents time**. Two distinct paths, one red and one green, visualize how a quantum particle travels through space-time.

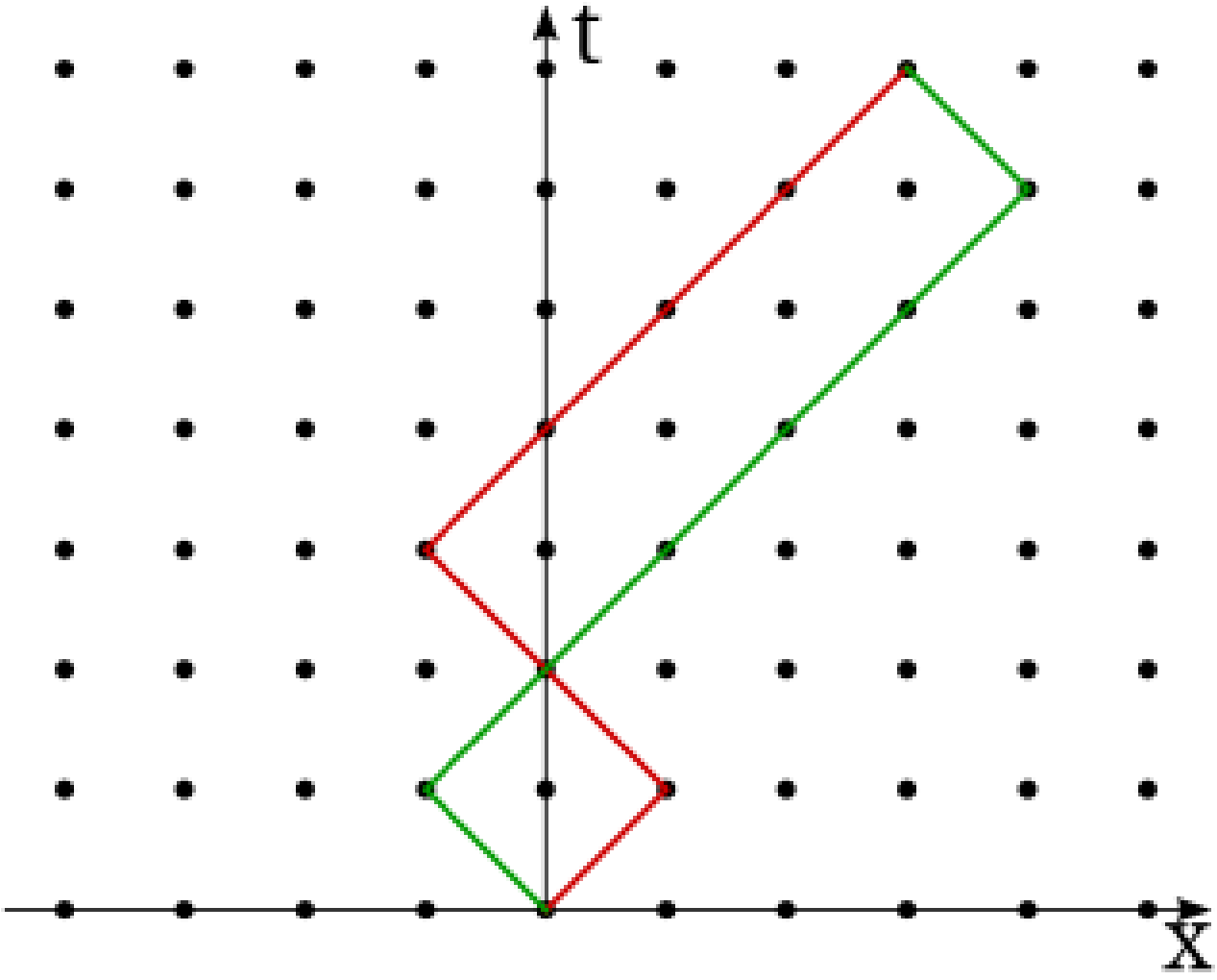
Both paths start at the exact same origin at the bottom (x=0, t=0). With every step upward in time, the particle must move exactly one unit diagonally left or right. Because the space step matches the time step, the particle travels at the speed of light, keeping all lines at a strict **45° angle**.

In quantum mechanics, a particle takes *every possible path at the same time*. This visualization highlights two alternative routes that happen to arrive at the exact same final destination at the top:

- **The Red Path:** Starts by moving right, makes a **rigid 90° turn** to the left, makes another **90° turn** to the right, and then runs straight to the end. It has a total of **2 turns**.
- **The Green Path:** Starts by moving left, makes a **90° turn**

to the right, travels straight for a long stretch, and makes a final **90° turn** to the left. It also has a total of **2 turns**.

Every time a line changes direction (a 90° bend), the internal quantum phase of the particle rotates. Mathematically, Feynman represents each turn by multiplying the particle's state by the complex factor $-\mathbf{i}$. Because both the red and green paths contain exactly **2 turns**, they undergo the same mathematical shift: $(-\mathrm{i})\ \mathrm{X}\ (-\mathrm{i})\ =\ -1$. Since they are perfectly in sync at the destination, their quantum waves constructively **reinforce each other** rather than cancelling out.



Symmetric Phase Domain.

According to the principles of **Schwarz** the 8x8 is transformed into a 15x15 square grid. For context the formula of Schwarz is $F(\bar{z})=\overline{F(z)}$. To translate this formula in human words, it is mirroring with preservation of internal logic. The more interesting thing is that the overall grid came out as a 15x15. That is logical in hindsight but it seemed notable since the point zero of this grid has the value of 15 (13GDS).

The 15x15 matrix represents a fully realized, discrete phase domain engineered using the principles of the Greek Duodecimal System (GDS). By applying the Schwarz Reflection Principle, the localized boundaries of a standard 8x8 chessboard are analytically extended into a four-quadrant, balanced field. This structural expansion does not merely repeat the numbers; it mirrors the mathematical relationships with perfect preservation of internal logic. Rather than dealing with an infinite, open coordinate system, this process constructs a highly organized, closed system where every coordinate corresponds to a precise geometric angle.

							←	0	→						
	80	α0	0	20	40	60	80	**α0**	80	60	40	20	0	α0	80	
	α0	13	46	79	ω0	23	56	**89**	56	23	ω0	79	46	13	α0	
	0	46	90	16	60	α6	30	**76**	30	α6	60	16	90	46	0	
	20	79	16	73	10	69	6	**63**	6	69	10	73	16	79	20	
	40	ω0	60	10	80	30	α0	**50**	α0	30	80	10	60	ω0	40	
	60	23	α6	69	30	ω3	76	**39**	76	ω3	30	69	α6	23	60	
↑	80	56	30	6	α0	76	50	**26**	50	76	α0	6	30	56	80	↑
0	**α0**	**89**	**76**	**63**	**50**	**39**	**26**	**13**	**26**	**39**	**50**	**63**	**76**	**89**	**α0**	**0**
↓	80	56	30	6	α0	76	50	**26**	50	76	α0	6	30	56	80	↓
	60	23	α6	69	30	ω3	76	**39**	76	ω3	30	69	α6	23	60	
	40	ω0	60	10	80	30	α0	**50**	α0	30	80	10	60	ω0	40	
	20	79	16	73	10	69	6	**63**	6	69	10	73	16	79	20	
	0	46	90	16	60	α6	30	**76**	30	α6	60	16	90	46	0	
	α0	13	46	79	ω0	23	56	**89**	56	23	ω0	79	46	13	α0	
	80	α0	0	20	40	60	80	**α0**	80	60	40	20	0	α0	80	
							←	0	→							

At the literal heart of this domain lies the central singularity, designated by the duodecimal value 13 (which evaluates to 15 in denary notation). This positioning means the entire matrix radiates from an absolute ground state of exactly 15 degrees. As one traces the values outward from this central node along the primary axes, a highly structured sequence emerges, stepping sequentially through the twelvefold divisions of a circle: 26 (30°), 39 (45°), 50 (60°), 63 (75°), 76 (90°), 89 (105°), and α0 (120°). The grid is not uniform; instead, the interaction of these numerical steps across the diagonal vectors generates concentric shells of values, highlighted by the color-coded rings. This structural configuration creates a standing wave or nodal interference pattern, mapping out localized geometric rules across the entire array.

The outermost boundary of the matrix, marked by the green perimeter zone, establishes the periodic boundary conditions necessary to keep the system mathematically stable. Every terminal value on this frame consists of a clean, uncorrupted multiple of 24, including 0, 20, 40, 60, 80, and α0. Operating under a Modulo-144 field collapse, these edges function as perfect mathematical mirrors. When a particle or localized wave vector reaches these peripheral zones, it does not escape or create a logical discontinuity. Instead, the boundary conditions cause the path to fold back seamlessly into the interior matrix, preserving the overall energy and configuration of the system.

This structural design introduces a critical evolution to the traditional Feynman Checkerboard Model. In Feynman's original formulation, a quantum particle moves along a boundless, flat space-time grid, where its internal clock shifts by a rigid 90 degrees only when changing direction.

By replacing that open space with this 15x15 Schwarz-extended matrix, the particle's movement becomes inherently bounded and guided by the geometry of the domain itself. Every single cell on this board contains a distinct, base-12 angular phase. As an electron pulse or vector operator steps from coordinate to coordinate, it continuously interacts with these pre-established local definitions. Consequently, calculating the probability of a path no longer depends solely on arbitrary turns, but becomes directly tethered to the underlying geometric framework of the matrix.

Defining the Static Phase Space and Boundary Conditions.

To establish a mathematically rigorous framework for this model, we must first separate the geometry of the space from the dynamics of the particles moving within it. We define our 15x15 matrix not as a passive game board, but as a discrete, static scalar field of geometric phases. This coordinate space operates entirely within the Greek Duodecimal System (GDS). Before any vector operator or quantum particle is introduced, the spatial dimensions, local phase values, and system boundaries must be explicitly defined according to the following criteria.

1. The Coordinate Space and Grid Resolution

We define a two-dimensional, discrete Cartesian coordinate system where the absolute centre of the matrix is designated as the origin, or zero-point, at coordinates (0,0). The horizontal and vertical axes extend symmetrically from this centre point across a domain of [-7, +7] along both the x and y dimensions, yielding a bounded matrix of exactly 15 X 15 discrete cells. Each spatial step of $\Delta x = 1$ or $\Delta y = 1$ does not merely represent a change in position; it corresponds mathematically to a fixed angular translation of exactly 15°. Consequently, distance within this grid is fundamentally unified with angular rotation.

2. The Local Phase Field and Schwarz Symmetry

Every individual cell within this coordinate space is pre-loaded with a static scalar value representing a specific angular phase. The ground state of the system is anchored at the origin (0,0) with a base value of 13 GDS (evaluating to 15° in denary notation). Rather than populating the grid arbitrarily, the full 15x15 field is constructed by taking a localized 8 X 8 sector and extending it outward across the axes via the Schwarz Reflection Principle. This analytic continuation ensures four-quadrant mirror symmetry, meaning that the internal mathematical logic is preserved identically across the entire domain, and the local field equations remain consistent from one quadrant to the next.

3. Boundary Conditions and Modulo-144 Confinement

The outermost perimeter of the matrix, located at the boundary lines where $x = \pm 7$ or $y = \pm 7$, establishes the strict physical limits of this closed system. This green boundary zone is engineered with precise periodic boundary conditions, consisting exclusively of clean, uncorrupted multiples of 24, such as 0, 20, 40, 60, 80, and α0. The edge of the matrix operates under a Modulo-144 field collapse constraint. Mechanically, these boundary conditions prevent any numerical divergence or logical discontinuity at the perimeter. The frame functions as a perfect mathematical mirror, forcing any incoming waveform or path vector to fold back into the interior matrix, thereby ensuring complete energy and phase confinement within the 15x15 field.

4. Particle Wave Dynamics and Phase Interaction

With the static properties of the matrix and its boundaries firmly established, we can define the fundamental rules governing particle dynamics within this domain. A quantum particle or electron pulse (such as the linear Pawn vector) is defined not as a localized point, but as a discrete wavefunction ψ (x, y) possessing both a spatial coordinate and an internal quantum phase.

As the particle propagates from cell to cell across the 15x15 field, its internal phase does not evolve in a vacuum. Instead, the particle's quantum clock continuously interacts with the pre-loaded local phase values of the background matrix. Every spatial step updates the particle's state by accumulating the angular value of the target coordinate, multiplied by the complex rotation operator $e^{i\pi/12}$.

Furthermore, the Modulo-144 boundary conditions actively dictate the global path integrals of the particle. When a wave vector strikes the green perimeter zone at x = ± 7 or y = ± 7, it encounters a perfect mathematical mirror composed of clean multiples of 24. Rather than escaping the system, the particle undergoes a discrete phase reflection (Schwarz reflection), folding its trajectory back into the interior matrix. This confinement causes the forward-propagating and reflected path integrals to constructively or destructively interfere, transforming the random walk of a standard Feynman model into a stable, self-contained standing wave pattern centred around the origin.

Conformal Polar Mapping and Phase Rotation.

To transition the system from a flat coordinate grid to a dynamic, fluid energy vortex, we utilize a conformal mapping transformation inspired by the **Riemann** Mapping Theorem. In a classical Cartesian system, spatial steps along the horizontal and vertical axes are treated independently of angular changes. However, by establishing that a single unit step Δ x = 1, Δ y = 1 inherently represents a fixed rotation of exactly 15°, the 15x15 Cartesian phase space can be bent into a continuous polar coordinate framework defined by radius r and angle θ.

$$\theta = n \cdot \frac{\pi}{12}$$

Under this transformation, the spatial properties of the matrix take on an absolute geometric logic. The structural concentric rings moving from the outer green perimeter toward the core represent physical distance r. Much like stable planetary orbits, every coordinate along a specific ring remains at a constant radius from the central singularity. Conversely, the angular slices wrapping around the circle represent the evolution of time or the continuous advancement of the quantum phase θ. Navigating step-by-step through the angles is mathematically identical to the ticking of a localized system clock or a rotation within the complex plane.

The mathematical operator driving this continuous transformation is the complex phase factor:

$$e^{i\pi/12}$$

Since π/12 radians matches the system's baseline constant of exactly 15°, this exponential function acts as the internal engine of the vortex. When a quantum particle (such as a diagonal Bishop vector)

propagates through this polar field, it acts as a gravitational accelerator. As the particle advances through successive time slices (θ), its trajectory is continuously multiplied by this complex rotation operator. This phase interaction bends the linear progression of the particle into an inward-tightening spiral. The forward momentum is smoothly redirected through the phase domain, causing the wavefunction to collapse into a fluid, self-correcting vortex that terminates perfectly at the King singularity anchored at the zero-point centre (0,0).

Matrix Wave Operators.

Within this discrete 15x15 phase domain, chess pieces stop being strategic game characters and are formally redefined as localized matrix operators, resonant nodes, and wave vector functions. Instead of executing game rules, each piece performs a specific thermodynamic, electromagnetic, or computational operation across the 15° static matrix layout. This shift transitions the framework from game strategy directly into discrete wave mechanics and electrotechnics.

- **The Pawn (The Linear Electron Pulse)**
    - **Trajectory:** Linear, orthogonal forward progression along the primary grid lines.
    - **System Function:** Represents a single discrete voltage pulse propagating through the matrix space. As the Pawn moves straight across the coordinates, it accumulates the localized phase changes of the static field. This progression steps across the field to systematically build up the initial linear potential distribution (voltage) across a single quadrant before interacting with the system boundaries.

- **The bishop (The Centripetal Vortex Accelerator)**
    - **Trajectory:** Pure diagonal execution intersecting the coordinates at exact 45° angles.

System Function: Acts as the primary driver of the centripetal vortex transformation. Moving along the diagonal vectors, the bishop's trajectory interfaces directly with the concentric harmonic colour rings. By multiplying its position by the complex phase factor $e^{i\pi/12}$ at each diagonal step, the bishop bends the linear potentials of the Pawns, driving a tightening, spiral phase-collapse directed inward toward the centre.

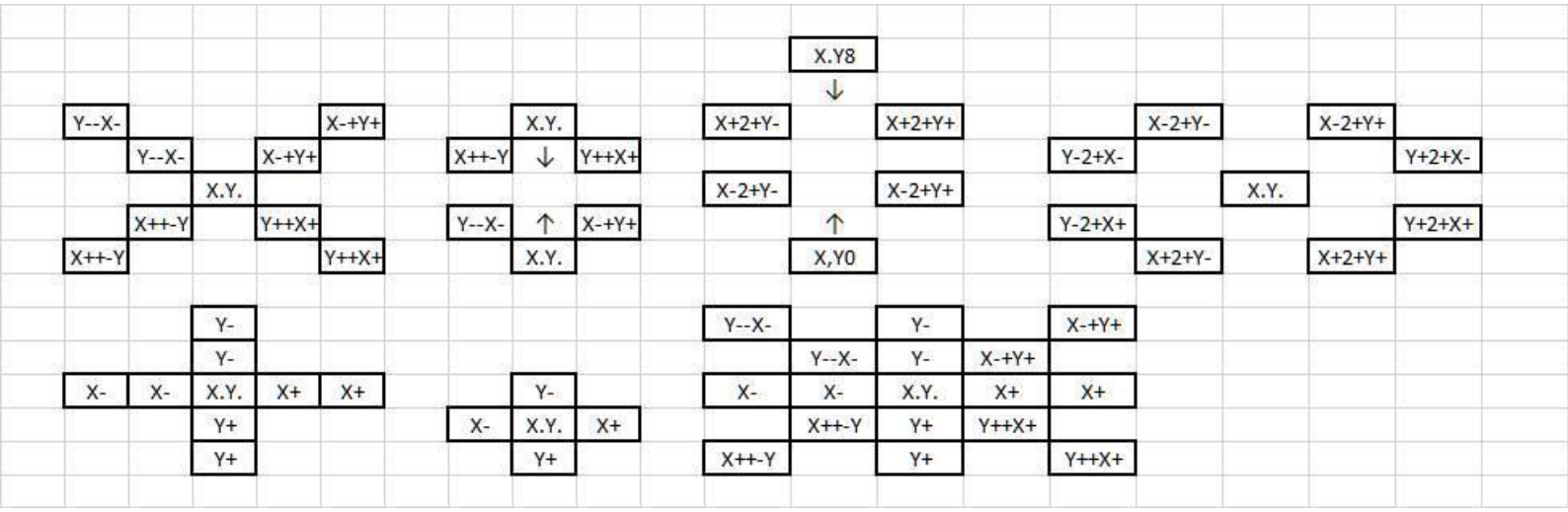
- **The Rook (The Transverse Waveguide)**
    - **Trajectory:** Infinite horizontal and vertical vector pathways parallel to the primary axes.
    - **System Function:** Governs the orthogonal boundaries and isolates the internal components of the field. The Rook guides transverse wave vectors along the primary axes, anchoring the phase distributions. By controlling the boundary lanes, the Rook locks the electromagnetic field profile in place, enforcing the Modulo-144 confinement parameters and preventing phase degradation outside the active array.

- **The Knight (The Inductive Phase Jumper)**
    - **Trajectory:** Asymmetric L-shaped vectors (2 steps forward, 1 step lateral), establishing non-local coordinate transformations.
    - **System Function:** Executes an instantaneous local phase-shift across the matrix. Geometrically, the Knight's discrete vector projects precise spatial angles of 26.56° or

63.43°. In this computational field, this movement represents the exact geometric window required for non-local wave tunnelling or wireless induction, allowing energy to shift between separated layers of the phase domain without executing continuous step-by-step translations through the intervening cells.

- **The Queen (The Omnidirectional Field Engine)**
  - **Trajectory:** Omnidirectional composition combining both orthogonal and diagonal vector pathways.
  - **System Function:** Represents the total, fluid electromagnetic current in high motion. The Queen synthesizes the linear, boundary-locking properties of the Rook with the spiral, phase-collapsing acceleration of the bishop. This dual action serves as a raw current engine, effortlessly driving complex field transformations across the entire 15x15 matrix layout.

- **The King (The Zero-Point Dielectric Singularity)**
  - **Trajectory:** Restricted single-step radius across all adjacent coordinates.
  - **System Function:** Anchors the absolute dielectric centre of the entire matrix configuration. Fixed at the zero-point origin (0,0) with the foundational base value of 13 GDS, the King remains virtually stationary as the unmoving axis of rotation. It acts as the ultimate sink for the balanced Modulo-144 field collapse, absorbing the converging phase vectors from the outer quadrants. If the balance at this central node is disrupted, the boundary conditions fail, causing the entire matrix configuration to collapse.



Like a gravitational field the rotation is time and the distance to the core is measurement. On a circular chessboard matrix, the traditional grid is bent into a polar system where the mechanics of physics take on a clear spatial logic.

The rings moving from the outside edge toward the centre represent physical distance. Just like being in orbit around a planet every position on a single ring remains at the exact same distance from the central singularity.

The slices wrapping around the circle represent the passage of time or the evolution of the quantum phase. Moving step-by-step through the angles is identical to the ticking of a clock or the rotation of the Earth.

When a particle moves through this matrix it experiences the local laws dictated by the chess pieces. The bishop acts as a gravitational accelerator bending linear motion into an inward spiral. As the particle rolls through the time slices it simultaneously falls through the spatial rings creating a fluid vortex that collapses perfectly toward the King at the zero-point centre.

Mathematical Derivations of the Matrix Wave Operators.

1. The Complex Rotation Engine

The foundational geometry of the Duodecimal Symmetric Phase Domain (DSPD) relies on a rigid coupling between spatial translation and angular transformation. A discrete spatial step of Δx=1 or Δy=1 inherently corresponds to a fixed rotation of 15° ($\pi/12$ radians) within the complex plane. The continuous phase evolution across the static scalar field is driven by the complex phase factor:

$$U = e^{i\pi/12}$$

Every discrete propagation step updates the localized particle wavefunction $\psi\,(x,y)$ by compounding this rotation engine against the background potentials.

2. The Pawn Operator (Linear Potential Field Generator)

The Pawn behaves as a linear, orthogonal vector function moving strictly along the primary grid lines. Its core systemic function is the step-by-step accumulation of localized phase transformations to construct a linear potential distribution (equivalent to a spatial voltage profile) across a single quadrant.

For a linear trajectory along a primary axis (such as the boundary row $Y1$), the voltage distribution V as a function of the discrete step n is defined by a linear modular translation:

$$V(n) = (\alpha 0 - 20n)\ (mod 144)$$

(Where α0=100 in the Greek Duodecimal System, establishing the starting boundary conditions).

3. The Bishop Operator (Centripetal Vortex Accelerator)

The bishop executes pure diagonal vectors intersecting the Cartesian coordinates at exact 45° angles. As it intercepts the concentric harmonic colour rings, it acts as a gravitational accelerator that warps linear potentials into an inward-tightening spiral phase-collapse.

By mapping the Cartesian coordinates into a conformal polar framework (z=x+iy=$r\cdot ei\theta$), the bishop wave operator continuously reduces the radius r while rotating the quantum phase θ:

$$\psi Bishop(Z_n) = r_n \cdot e^{i(\theta_0 + n\frac{\pi}{12})}$$

This drives the localized wave vector along a trajectory where the angular velocity collapses exponentially toward the central singularity.

4. The Knight Operator (Non-Local Inductive Phase Jumper)

The Knight executes asymmetric L-shaped spatial steps (±2, ±1), establishing instantaneous non-local coordinate transformations across the domain. Because its discrete displacement vector projects precise goniometric angles:

$$\theta 1 = tan^{-1}\left(\frac{1}{2}\right) \approx 26.56°$$
$$\theta 2 = tan^{-1}\left(\frac{2}{1}\right) \approx 63.43°$$

These trajectories operate outside the standard twelvefold 15° phase-grid intervals. Consequently, the Knight function bypasses local background interactions, acting as a discrete wave tunnelling operator. It maps energy instantaneously between separated layers of the phase domain without continuous step-by-step translation through intervening cells:

$$\mathcal{O}_{Knight}\psi(x,y) = \psi(x \pm 2, y \pm 1) \cdot e^{\mathrm{i}\phi\mathrm{tunnel}}$$

5. The King Operator (Zero-Point Dielectric Singularity)

The King acts as a static boundary condition anchored at the absolute centre of the coordinate system (0, 0). It is fixed at the foundational ground state value of 13^{GDS} (15° denary) and functions as a non-divergent Dirac-delta sink for the global Modulo-144 field collapse.
The structural equilibrium of the entire 15x15 field requires that the total influx of converging phase vectors from the outer quadrants matches the central boundary value:

$$\oint_{\partial Core} \psi(x,y)dl = \psi(0,0) = 13^{GDS}$$

If this central energy sink is perturbed, the periodic boundary conditions destabilize, causing an immediate phase degradation and structural collapse of the active array.

6. The Rook Operator (Transverse Waveguide)

The Rook functions as a transverse waveguide vector operator, restricted to infinite horizontal and vertical pathways parallel to the primary grid axes (Δx=0 or Δy=0). Its systemic function is to enforce boundary isolation and stabilize the electromagnetic field profile across the coordinate network.
By guiding wave vectors strictly along the orthogonal axes, the Rook locks the phase distribution into a rigid cross-symmetrical framework. Operating under the Modulo-144 field collapse constraint, the Rook operator acts as an orthogonal projector $\mathcal{P}$ that prevents phase degradation beyond the active array by mapping non-boundary leakage vectors back into the localized grid coordinates:

$$\mathcal{O}_{Rook}\psi(x,y) = \mathcal{P}_{orthogonal}[\psi(x,y)](mod\ 144)$$

This stabilizing interaction structurally locks the standing wave nodes in place, maintaining the rigid internal grid logic required for geometric confinement.

7. The Queen Operator (Omnidirectional Field Engine)

The Queen behaves as an omnidirectional field engine, combining the orthogonal boundaries of the Rook with the exponential spiral acceleration of the Bishop. In this computational domain, the Queen represents the total, fluid electromagnetic current in high motion.
Mathematically, the Queen operator is a composite wave function that synthesizes linear potential expansion with centripetal phase-collapse. It scans across all adjacent vectors simultaneously, executing a superposition of both Cartesian translations and complex polar rotations:

$$\mathcal{O}_{Queen}\psi(z) = (\mathcal{O}_{Rook} + \mathcal{O}_{Bishop})\psi(z)$$

By evaluating both linear potential gradients and exponential phase factors ($e^{i\pi/12}$) across all vectors, the Queen serves as the raw current driver that effortlessly translates and mutates complex field configurations across the entire 15×15 matrix layout.

Discrete Time-Evolution and Multi-Operator Dynamics.

While the 15×15 matrix defines a static scalar field of spatial phases, a dynamic quantum system requires a formal definition of time-evolution. In this discrete computational domain, time does not progress continuously, but advances in unitary increments of Δt=1, equivalent to a sequential "turn" or system tick.

The global quantum state of the board at any given time step t is represented by the total time-dependent wavefunction $\Psi(x,y,t)$. To transition the system from state t to t+1, we introduce a discrete Time-Evolution Operator, $\mathcal{U}(\Delta t)$. This operator computes the multi-operator superposition of all active piece vectors currently intersecting the phase space:

$$\Psi(\mathrm{x},\mathrm{y},\mathrm{t}+1) = \mathcal{U}(\Delta\mathrm{t})\Psi(\mathrm{x},\mathrm{y},\mathrm{t})$$

Where $\mathcal{U}$ (Δt) is the sum of all local matrix operators $\mathcal{O}_k$ active on the board, weighted by their spatial coordinates and phase constraints:

$$\mathcal{U}(\Delta\mathrm{t}) = \exp\left(\sum_k \mathcal{O}_k \cdot \Delta\mathrm{t}\right)$$

When multiple pieces move or exert influence simultaneously, their wavefunctions do not merely add up linearly. Instead, they generate dynamic phase interference patterns across the concentric colour rings. The forward-propagating potentials and boundary-reflected waveforms (governed by the Modulo-144 constraint) collapse and reconstruct at each tick of the system clock. This transforms the static grid into a living, fluid energy vortex where the historical paths of all pieces actively dictate the local probability density of the next state.

This document is presented as a conceptual whitepaper and a theoretical framework rather than a concrete, experimentally validated scientific discovery. The structures outlined herein map the boundary limits of my current geometric exploration, bridging the gap between discrete topological matrix models and wave mechanics. It is intended to serve as a foundational blueprint and an open invitation for computational physicists and software engineers to expand, simulate, and test these base-12 phase domain transformations.

MadHatter out.

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